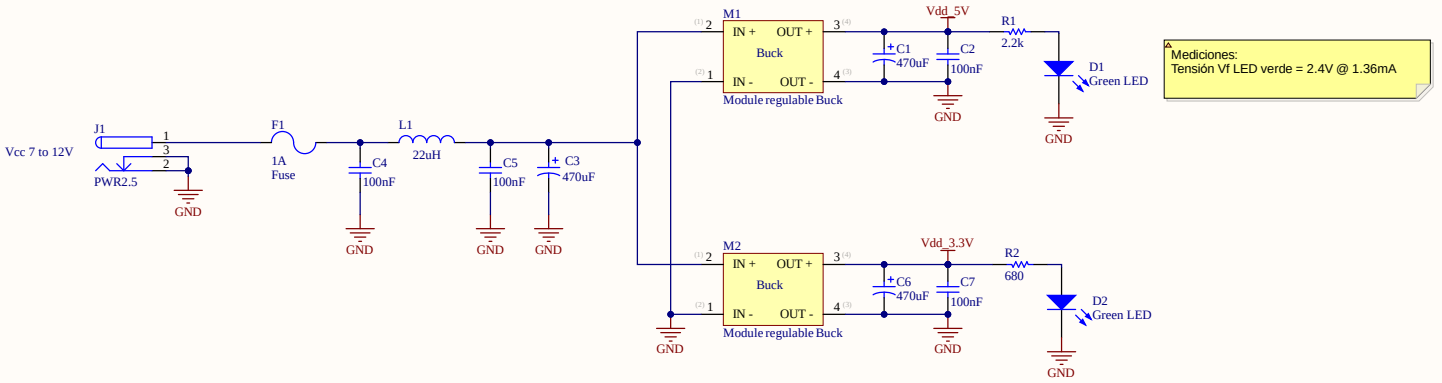
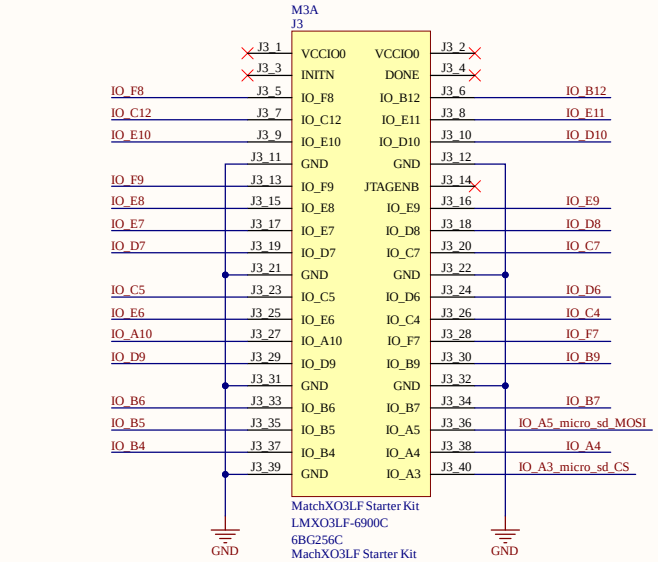
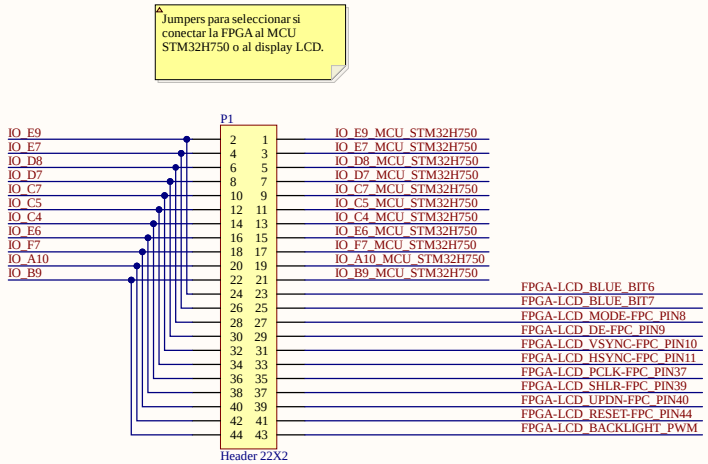
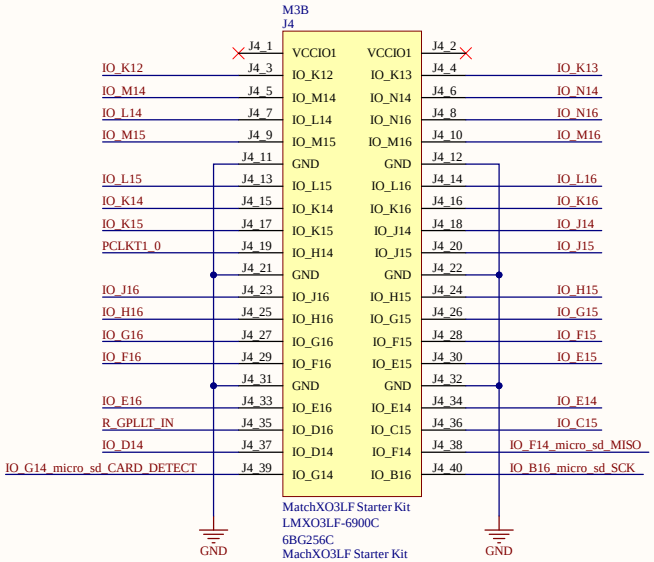


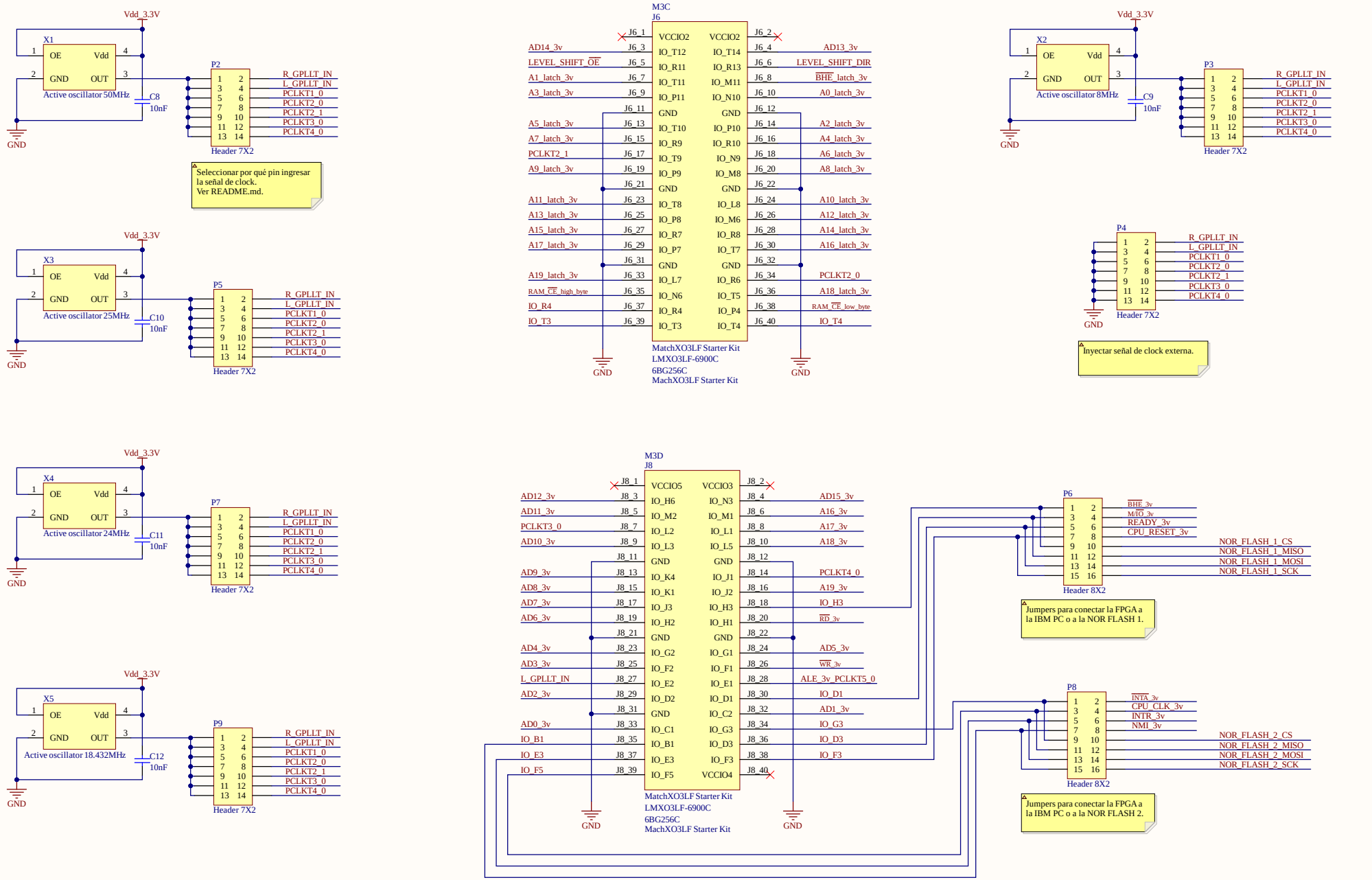
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Size A3	Number	Revision
Date: 10/17/2026		Sheet of
File: C:\Users\..\power_supply.SchDoc		Drawn By:





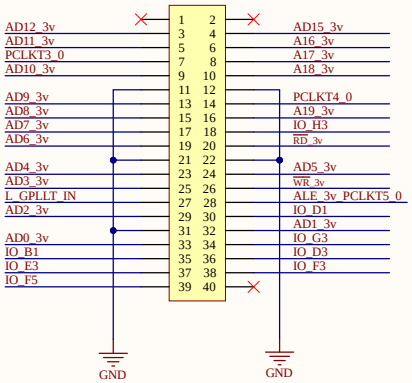


Title		
Size A3	Number	Revision
Date: 10/17/2026		Sheet of
File: C:\Users\..\FPGA_J3_J4.SchDoc		Drawn By:

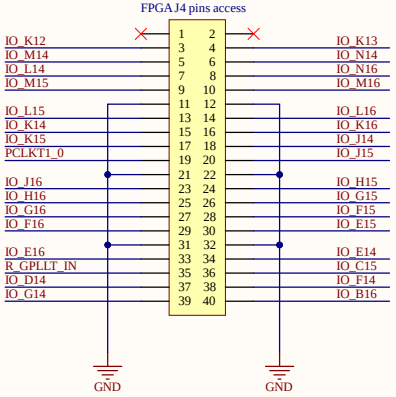

Vdd_3.3V
X1
OE
Vdd
GND
OUT
Active oscillator 50MHz
C8
10nF
GND
P2
Header 7X2
R_GPLLT_IN
L_GPLLT_IN
PCLKT1_0
PCLKT2_0
PCLKT2_1
PCLKT3_0
PCLKT4_0
Seleccionar por qué pin ingresar la señal de clock.
Ver README.md.
X3
Active oscillator 25MHz
C10
P5
X4
Active oscillator 24MHz
C11
P7
X5
Active oscillator 18.432MHz
C12
P9
M3C
J6
VCCIO2
IO_T12
IO_R11
IO_T11
IO_P11
IO_T10
IO_R9
IO_T9
IO_P9
IO_T8
IO_P8
IO_R7
IO_P7
IO_L7
IO_N6
IO_R4
IO_T3
IO_T14
IO_R13
IO_M11
IO_N10
IO_P10
IO_R10
IO_N9
IO_M8
IO_L8
IO_M6
IO_R8
IO_T7
IO_R6
IO_T5
IO_P4
IO_T4
AD14_3v
LEVEL_SHIFT_OE
A1_latch_3v
A3_latch_3v
A5_latch_3v
A7_latch_3v
PCLKT2_1
A9_latch_3v
A11_latch_3v
A13_latch_3v
A15_latch_3v
A17_latch_3v
A19_latch_3v
RAM_CE_high_byte
IO_R4
IO_T3
AD13_3v
LEVEL_SHIFT_DIR
BHE_latch_3v
A0_latch_3v
A2_latch_3v
A4_latch_3v
A6_latch_3v
A8_latch_3v
A10_latch_3v
A12_latch_3v
A14_latch_3v
A16_latch_3v
PCLKT2_0
A18_latch_3v
RAM_CE_low_byte
IO_T4
MatchXO3LF Starter Kit
LMXO3LF-6900C
6BG256C
MachXO3LF Starter Kit
X2
Active oscillator 8MHz
C9
P3
P4
Inyectar señal de clock externa.
M3D
J8
VCCIO5
VCCIO3
IO_H6
IO_N3
IO_M2
IO_M1
IO_L2
IO_L1
IO_L3
IO_L5
IO_K4
IO_J1
IO_K1
IO_J2
IO_J3
IO_H3
IO_H2
IO_H1
IO_G2
IO_G1
IO_F2
IO_F1
IO_E2
IO_E1
IO_D2
IO_D1
IO_C2
IO_C1
IO_G3
IO_B1
IO_D3
IO_E3
IO_F3
IO_F5
VCCIO4
AD12_3v
AD11_3v
PCLKT3_0
AD10_3v
AD9_3v
AD8_3v
AD7_3v
AD6_3v
AD4_3v
AD3_3v
L_GPLLT_IN
AD2_3v
AD0_3v
IO_B1
IO_E3
IO_F5
AD15_3v
A16_3v
A17_3v
A18_3v
PCLKT4_0
A19_3v
IO_H3
RD_3v
AD5_3v
WR_3v
ALE_3v_PCLKT5_0
IO_D1
AD1_3v
IO_G3
IO_D3
IO_F3
P6
BHE_3v
M/IO_3v
READY_3v
CPU_RESET_3v
NOR_FLASH_1_CS
NOR_FLASH_1_MISO
NOR_FLASH_1_MOSI
NOR_FLASH_1_SCK
Header 8X2
Jumpers para conectar la FPGA a la IBM PC o a la NOR FLASH 1.
P8
INTA_3v
CPU_CLK_3v
INTR_3v
NMI_3v
NOR_FLASH_2_CS
NOR_FLASH_2_MISO
NOR_FLASH_2_MOSI
NOR_FLASH_2_SCK
Jumpers para conectar la FPGA a la IBM PC o a la NOR FLASH 2.
Title
Size
A3
Number
Revision
Date:
10/17/2026
File:
C:\Users\..\FPGA_J6_J8.SchDoc
Sheet of
Drawn By:

Estos pines son para acceder a las señales con instrumentos como el analizador lógico y el osciloscopio.
También sirven para conectar placas de pruebas.

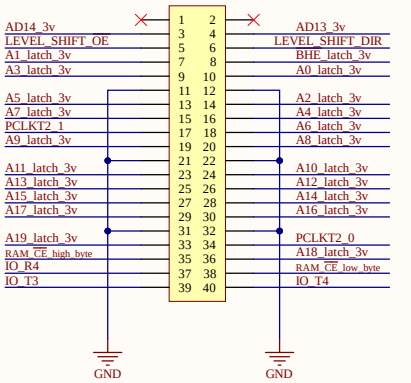
P11
FPGA J8 pins access



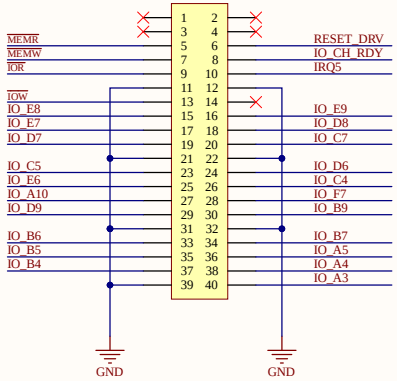
P10
FPGA J4 pins access



P13
FPGA J6 pins access

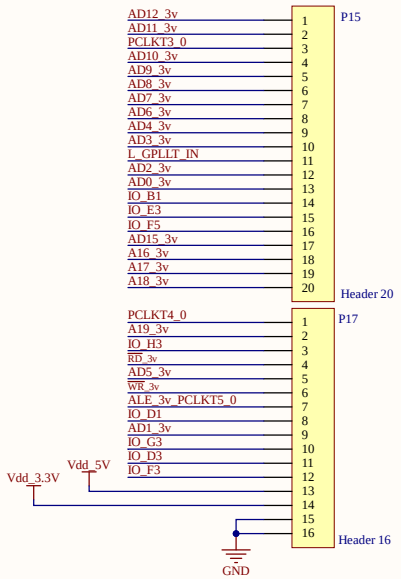


P12
FPGA J3 pins access



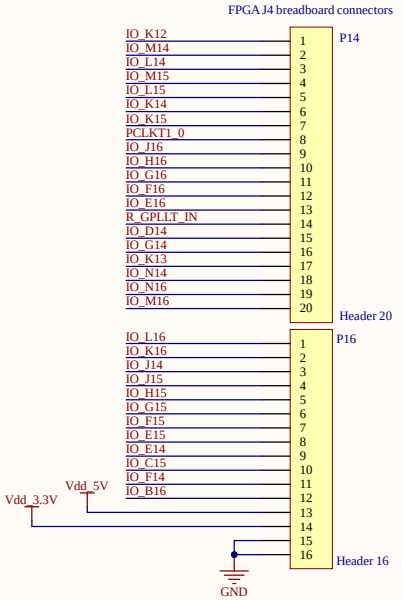
Title			
Size: A3	Number		Revision
Date: 10/17/2026		Sheet of	
File: C:\Users\..\FPGA_pin_access.SchDoc		Drawn By:	

FPGA J8 breadboard connectors

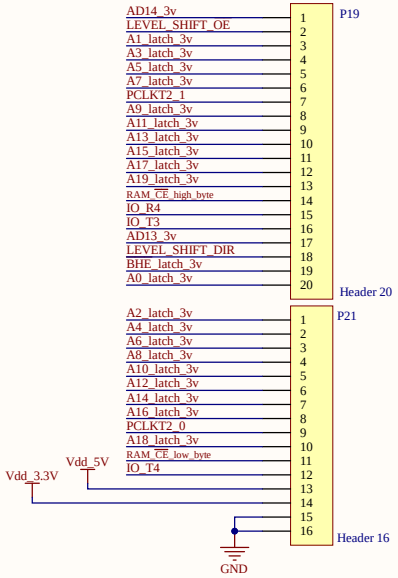


Estos pines son para conectar los pines de la FPGA a breadboards externas.

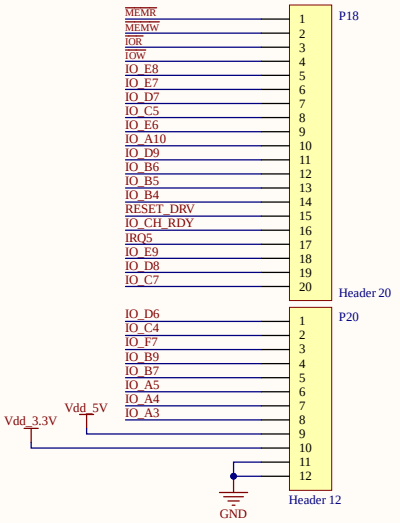
FPGA J4 breadboard connectors



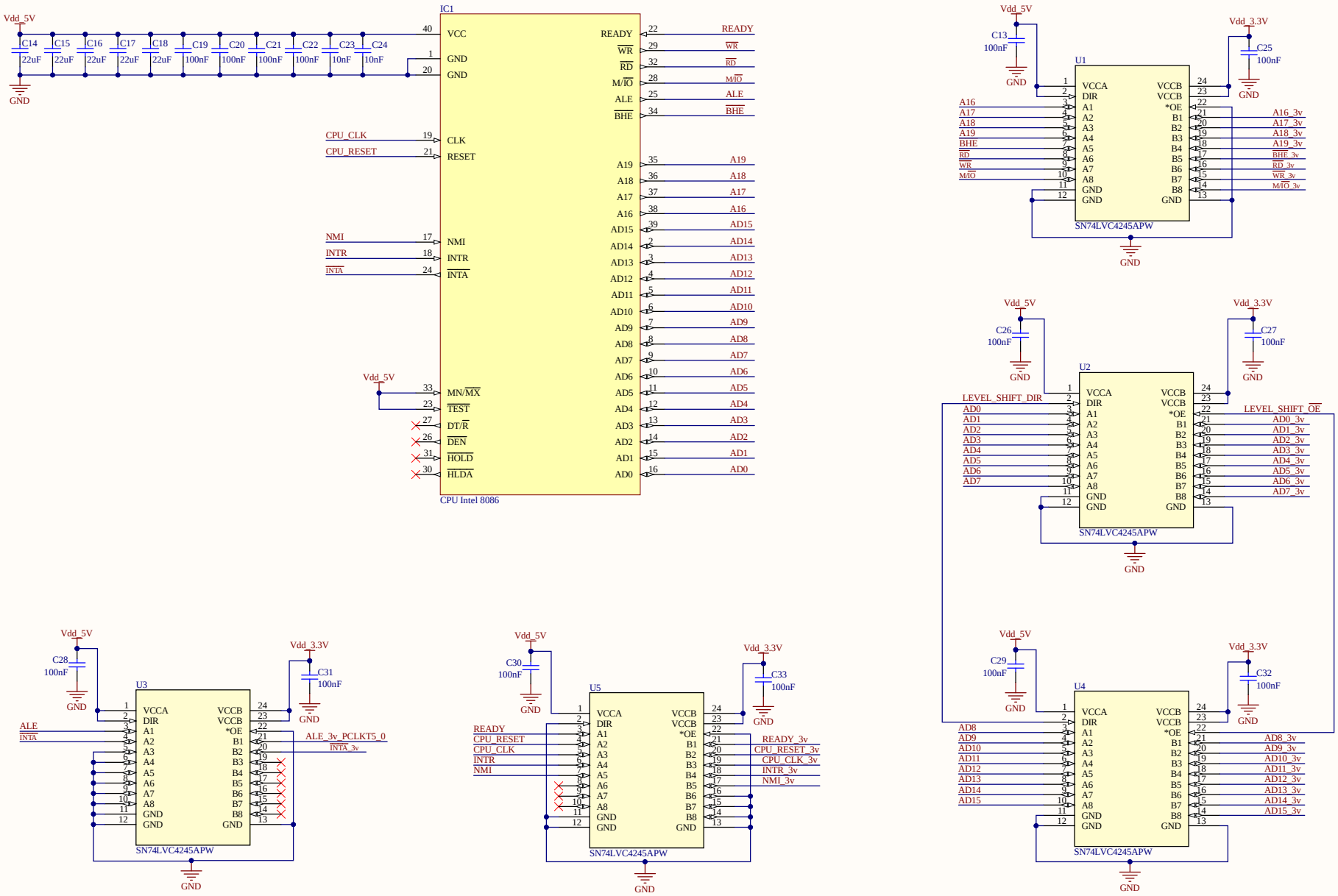
FPGA J6 breadboard connectors



FPGA J3 breadboard connectors

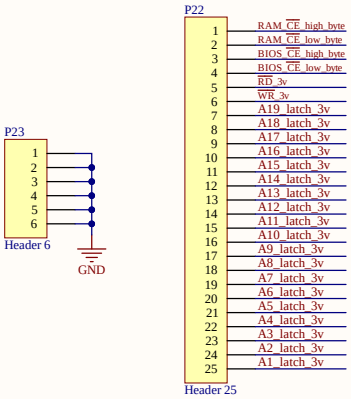


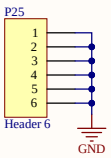
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Size: A3	Number	Revision
Date: 10/17/2026	Sheet of	
File: C:\Users\..\FPGA_breadboard_connections.SchDoc	Drawn By:	

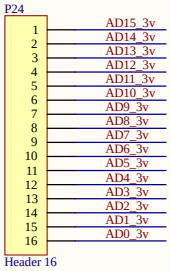


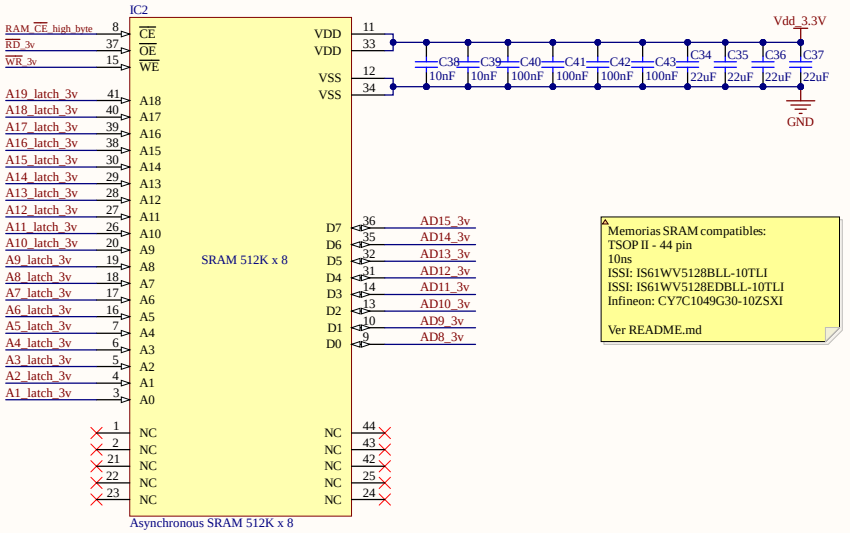
Title		
Size A3	Number	Revision
Date: 10/17/2026		Sheet of
File: C:\Users\..\IBM_PC_CPU.SchDoc		Drawn By:

Acceso a los buses de direcciones y datos para su monitorización mediante un analizador lógico.



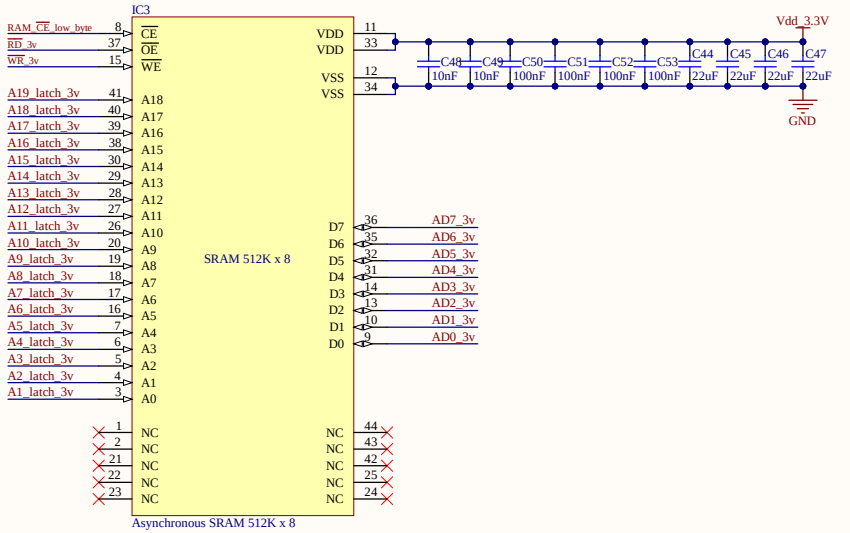






Memorias SRAM compatibles:
TSOP II - 44 pin
10ns
ISSI: IS61WV5128BLL-10TLI
ISSI: IS61WV5128EDBLL-10TLI
Infineon: CY7C1049G30-10ZSXI

Ver README.md

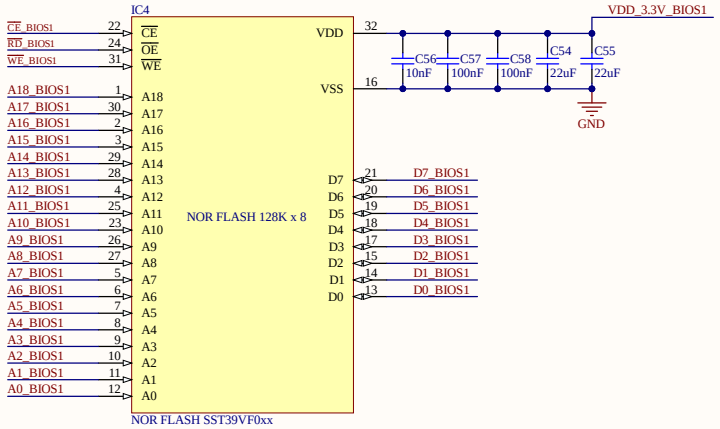


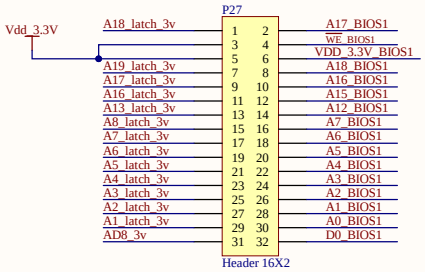
Title		
Size: A3	Number	Revision
Date: 10/17/2026		Sheet of
File: C:\Users\..\IBM_PC_RAM.SchDoc		Drawn By:

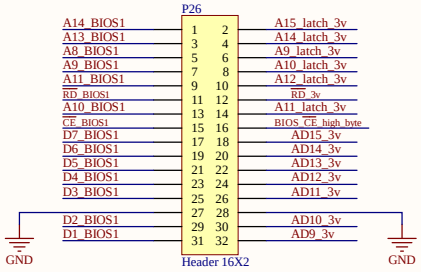
Para posicionar la BIOS en determinado rango de direcciones en el mapa de memoria, la FPGA primero lee la dirección enviada por la CPU y luego usa las señales CE* de cada chip para seleccionarlos.

Chips compatibles:
SST39VF010-70-4I-NHE-T
SST39VF010-55-4I-NHE-T
SST39VF020-70-4I-NHE-T
SST39VF020-55-4I-NHE-T
SST39VF040-70-4I-NHE-T
SST39VF040-55-4I-NHE-T

Si se instalan chips SST39VF040, no es necesario direccionar los pines 1 y 30.

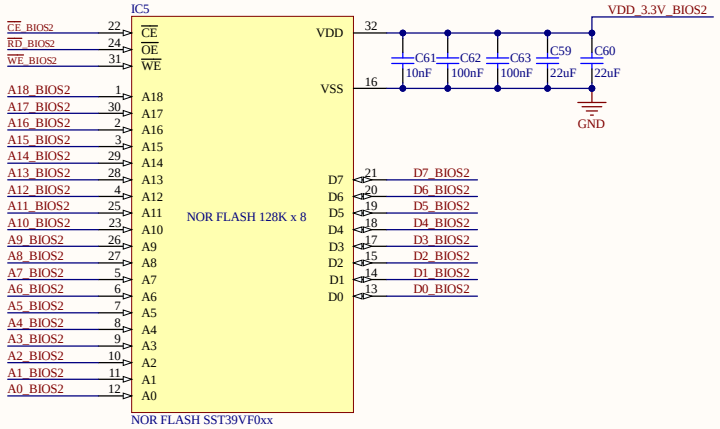


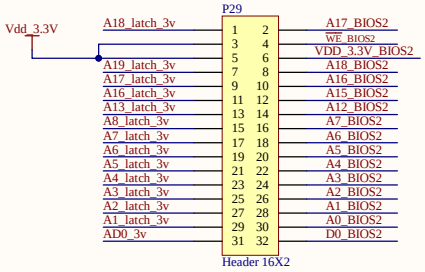


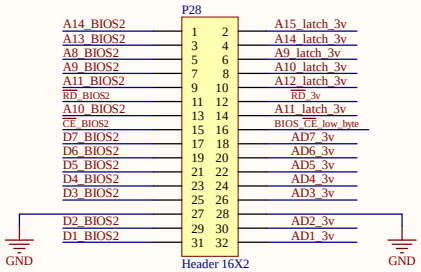


Jumpers para desconectar los chips de BIOS de la placa base y conectarlos a un programador así no es necesario extraer los chips del zócalo para programarlos.

Jumpers para desconectar los chips de BIOS de la placa base y conectarlos a un programador así no es necesario extraer los chips del zócalo para programarlos.

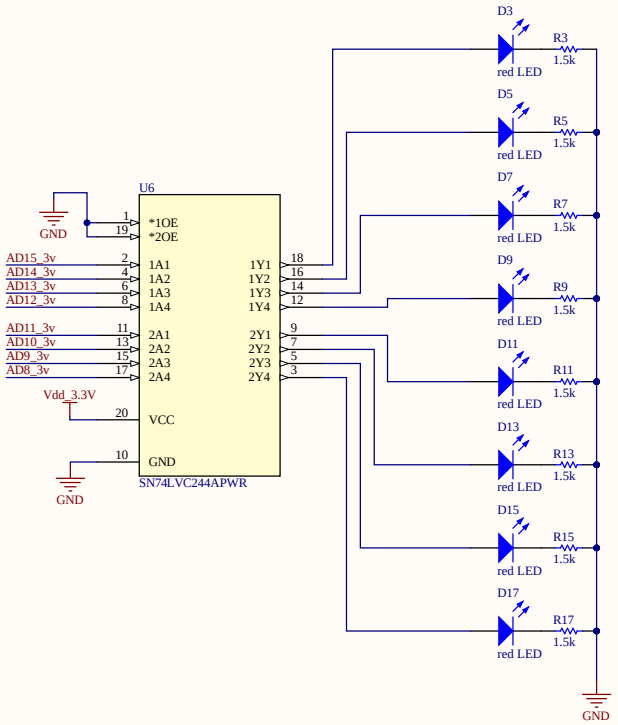


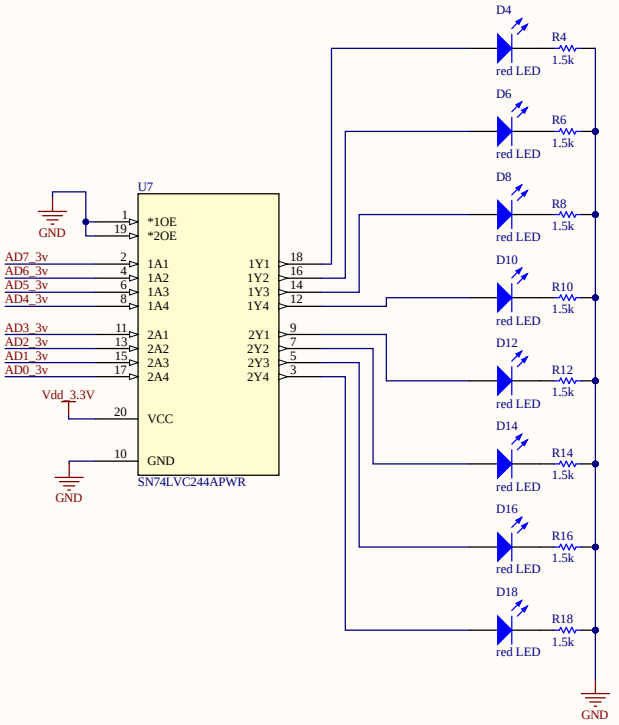




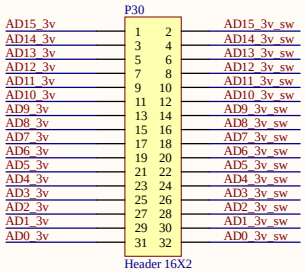
Title		
Size A3	Number	Revision
Date: 10/17/2026	Sheet of	
File: C:\Users\..\IBM_PC_ROM_BIOS.SchDoc	Drawn By:	

Mediciones:
Tensión Vf LED rojo = 1.816V @ 1.2216mA.
La intensidad del LED rojo 0603 con 1.2216mA es óptima, no molesta a la vista y se distingue muy bien.

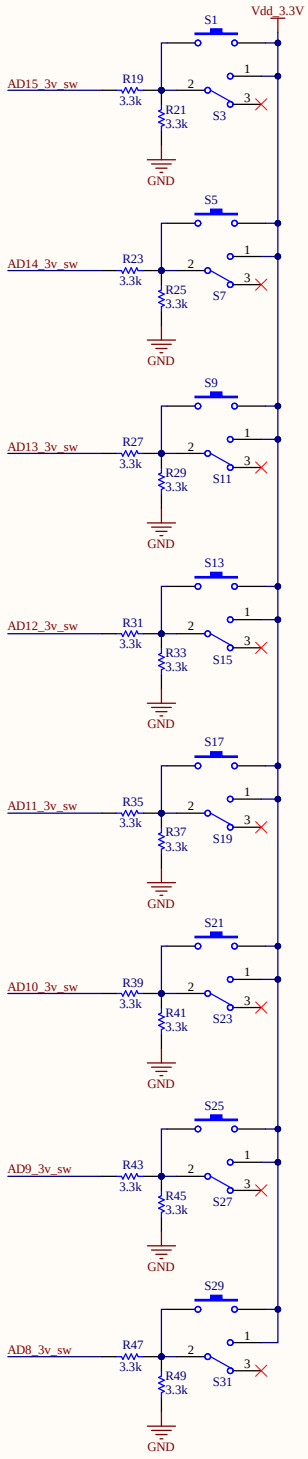


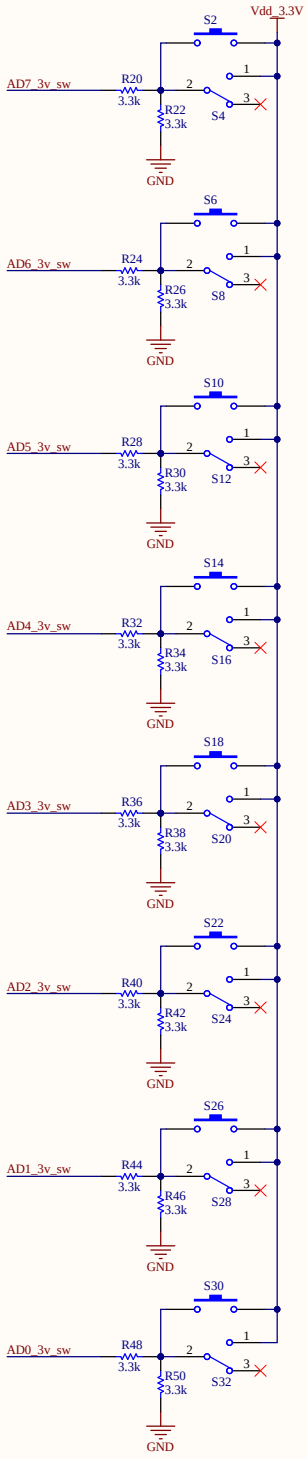


Title		
Size A3	Number	Revision
Date: 10/17/2026		Sheet of
File: C:\Users\..\pc_data_bus_leds.SchDoc		Drawn By:

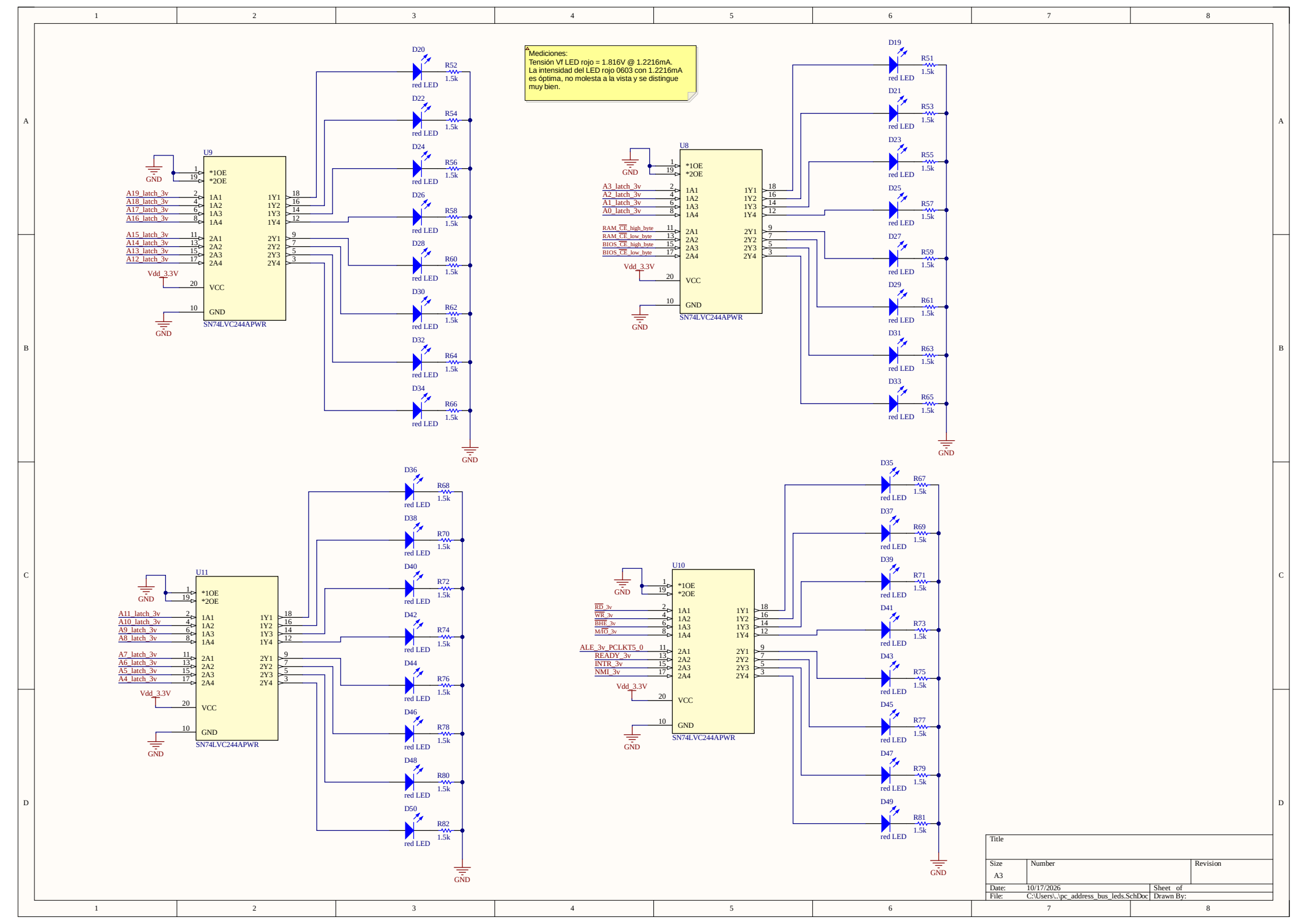


Jumpers para desconectar circuitos de entradas si no se utilizan, para no alterar otras señales.





Title		
Size: A3	Number	Revision
Date: 10/17/2026	Sheet of	
File: C:\Users\..\pc_data_bus_input_switches.SchDoc	Drawn By:	


Mediciones:
Tensión Vf LED rojo = 1.816V @ 1.2216mA.
La intensidad del LED rojo 0603 con 1.2216mA es óptima, no molesta a la vista y se distingue muy bien.
U9
U8
U11
U10
SN74LVC244APWR
*1OE
*2OE
1A1
1A2
1A3
1A4
2A1
2A2
2A3
2A4
1Y1
1Y2
1Y3
1Y4
2Y1
2Y2
2Y3
2Y4
VCC
GND
Vdd_3.3V
A19_latch_3v
A18_latch_3v
A17_latch_3v
A16_latch_3v
A15_latch_3v
A14_latch_3v
A13_latch_3v
A12_latch_3v
A3_latch_3v
A2_latch_3v
A1_latch_3v
A0_latch_3v
RAM_CE_high_byte
RAM_CE_low_byte
BIOS_CE_high_byte
BIOS_CE_low_byte
A11_latch_3v
A10_latch_3v
A9_latch_3v
A8_latch_3v
A7_latch_3v
A6_latch_3v
A5_latch_3v
A4_latch_3v
RD_3v
WR_3v
BHE_3v
M/IO_3v
ALE_3v_PCLKT5_0
READY_3v
INTR_3v
NMI_3v
D20
D22
D24
D26
D28
D30
D32
D34
D19
D21
D23
D25
D27
D29
D31
D33
D36
D38
D40
D42
D44
D46
D48
D50
D35
D37
D39
D41
D43
D45
D47
D49
R52
R54
R56
R58
R60
R62
R64
R66
R51
R53
R55
R57
R59
R61
R63
R65
R68
R70
R72
R74
R76
R78
R80
R82
R67
R69
R71
R73
R75
R77
R79
R81
red LED
1.5k
Title
Size
A3
Number
Revision
Date:
10/17/2026
Sheet of
File:
C:\Users\..\pc_address_bus_leds.SchDoc
Drawn By:

Mediciones para luminosidad adecuada:

Vcc = 3.3V
R = 680 ohm o 820 ohm

Tensión de LED:
Vf = 1.866V @ 680 ohm
Vf = 1.852V @ 820 ohm

Corriente:
If = (3.3V - 1.866V) / 680 ohm = 2.109mA
If = (3.3V - 1.852V) / 820 ohm = 1.766mA
If = (3.3V - 1.87) / 560 ohm = 2.55mA

Se usan R de 560 ohm porque al multiplexar el encendido, baja la luminosidad de los segmentos.

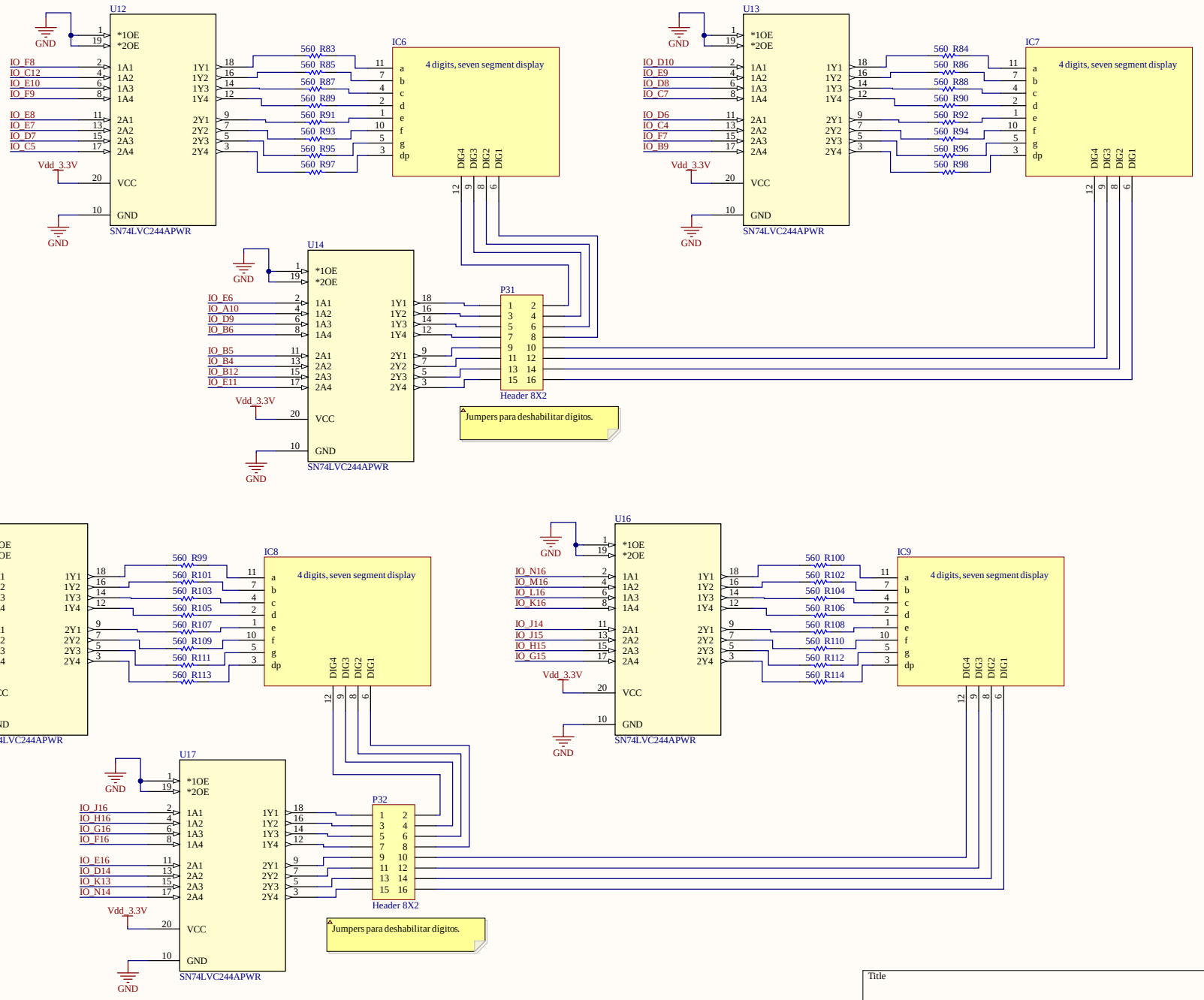
Cada chip driver debe entregar 20.4mA máximo con todos los segmentos encendidos y el máximo recomendado para un pin es de 24mA por lo que estamos dentro del límite.
El driver que controla el encendido de cada dígito de entregar 40.8mA máximo.

pin 6 a VDD selecciona el primer dígito de derecha a izquierda (el menos significativo).
Seleccionando el primer dígito, el pin 3 a GND no enciende nada.

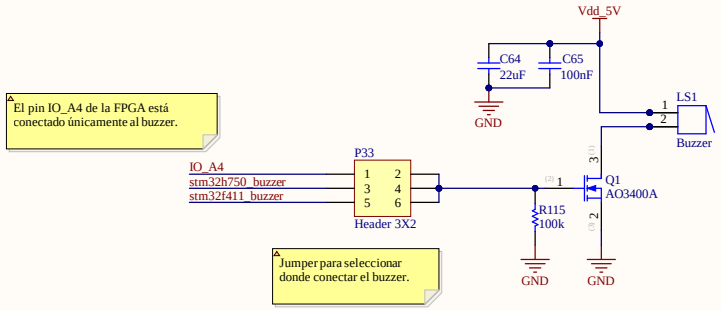
pin 8 a VDD selecciona el segundo dígito de derecha a izquierda.
Seleccionando el segundo dígito, el pin 3 a GND no enciende nada.

pin 9 a VDD selecciona el tercer dígito de derecha a izquierda.
pin 3 a GND con DIG2 a Vcc enciende los dos puntos de horas/minutos.

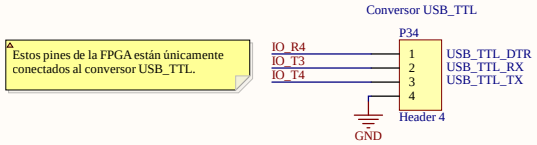
pin 12 a VDD selecciona el cuarto dígito de derecha a izquierda (el más significativo).
Seleccionando el cuarto dígito, el pin 3 a GND enciende el tercer punto de derecha a izquierda.



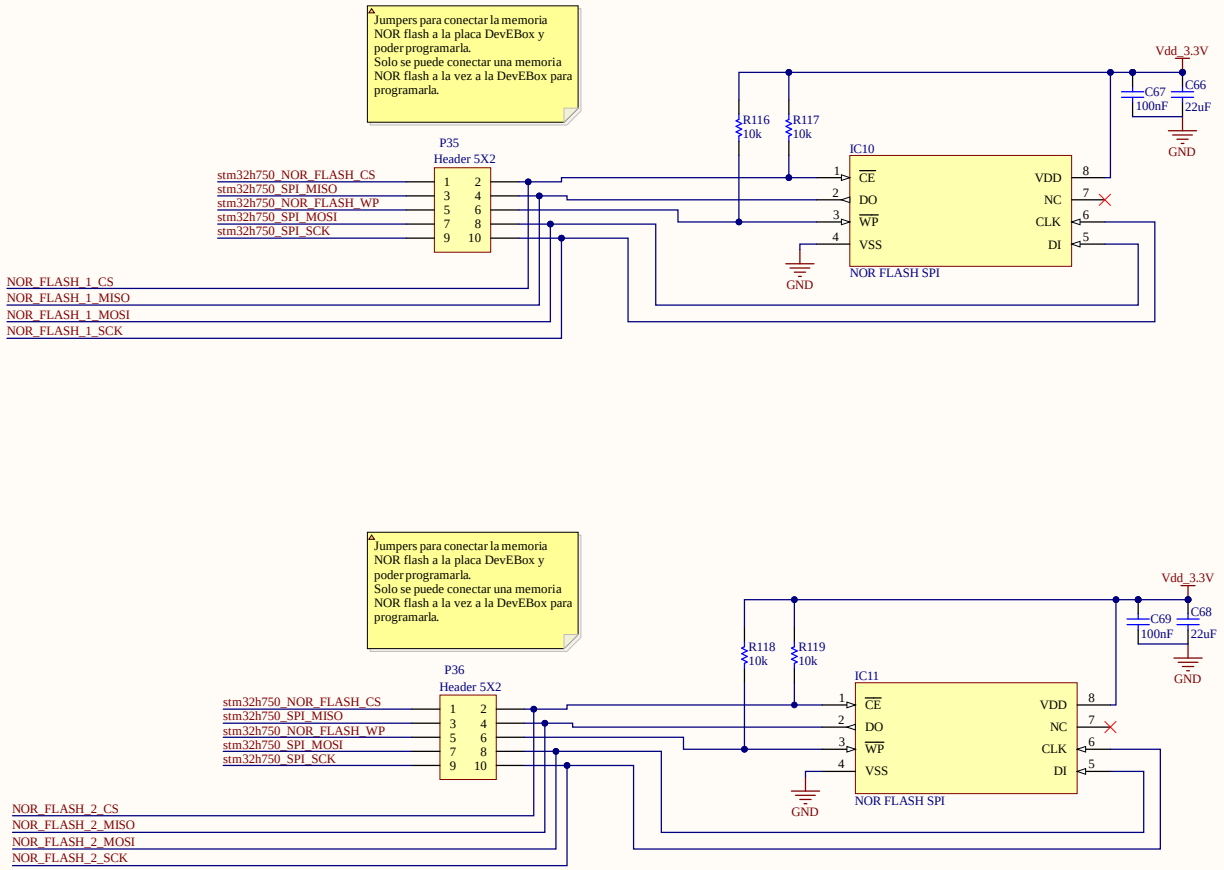
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Date: 10/17/2026	Sheet of	
File: C:\Users\..\seven_segment_displays.SchDoc	Drawn By:	



Title		
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Date: 10/17/2026		Sheet of
File: C:\Users\..\buzzer.SchDoc		Drawn By:



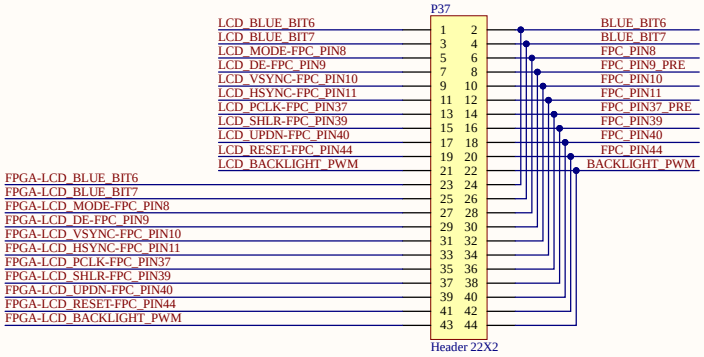
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File: C:\Users\..\USB_TTL_converter.SchDoc		Drawn By:	



Title		
Size A3	Number	Revision
Date: 10/17/2026		Sheet of
File: C:\Users\..\NOR_flash.SchDoc		Drawn By:


Vdd_5V
C70
100nF
GND
M4A
J2
5V
3.3V
GND
PD9
PD11
PD13
PD15
PC7
PC9
PA9/TXD1
PA11/USB_DM
PA15
PC11
PD0
PD2
PD4
PD6
PB3
PB5
PB7
PB9
PE1
5V
3.3V
GND
PD8
PD10
PD12
PD14
PC6
PC8
PA8
PA10/RXD1
PA12/USB_DP
PC10
PC12
PD1
PD3
PD5
PD7
PB4
PB6
PB8
PE0
J2_1
J2_3
J2_5
J2_7
J2_9
J2_11
J2_13
J2_15
J2_17
J2_19
J2_21
J2_23
J2_25
J2_27
J2_29
J2_31
J2_33
J2_35
J2_37
J2_39
J2_41
J2_43
J2_2
J2_4
J2_6
J2_8
J2_10
J2_12
J2_14
J2_16
J2_18
J2_20
J2_22
J2_24
J2_26
J2_28
J2_30
J2_32
J2_34
J2_36
J2_38
J2_40
J2_42
J2_44
LCD_RESET-FPC_PIN44
AD11_3v
AD13_3v
ALE_3v_PCLKT5_0
AD15_3v
WR_3v
LCD_UPDN-FPC_PIN40
IO_E3
IO_F8
IO_C12
IO_E10
IO_F9
IO_E8
IO_E7_MCU_STM32H750
IO_D7_MCU_STM32H750
IO_C5_MCU_STM32H750
IO_E6_MCU_STM32H750
LCD_BLUE_BIT7
IO_A10_MCU_STM32H750
LCD_MODE-FPC_PIN8
AD10_3v
AD12_3v
AD14_3v
LCD_HSYNC-FPC_PIN11
RD_3v
LCD_BACKLIGHT_PWM
LCD_SHLR-FPC_PIN39
IO_B12
IO_E11
IO_D10
IO_E9_MCU_STM32H750
IO_D8_MCU_STM32H750
IO_C7_MCU_STM32H750
stm32h750_NOR_FLASH_WP
IO_C4_MCU_STM32H750
IO_F7_MCU_STM32H750
LCD_BLUE_BIT6
IO_B9_MCU_STM32H750
Module DevEBox STM32H7XX_M
V2.0
SN: 1907
Module DevEBox STM32H7XX_M
Vdd_5V
C71
100nF
GND
M4B
J3
3.3V
GND
VDDA
PB15
PB13
PB11
PE15
PE13
PE11
PE9
PE7
PB1
PC5
PA7
PA5
PA3
PA1
PC3
PC1
PC13
PE5
PE3
3.3V
GND
VREF+
PB14
PB12
PB10
PE14
PE12
PE10
PE8
PB2
PB0
PC4
PA6
PA4
PA2
PA0
PC2
PC0
PE6
PE4
PE2
J3_1
J3_3
J3_5
J3_7
J3_9
J3_11
J3_13
J3_15
J3_17
J3_19
J3_21
J3_23
J3_25
J3_27
J3_29
J3_31
J3_33
J3_35
J3_37
J3_39
J3_41
J3_43
J3_2
J3_4
J3_6
J3_8
J3_10
J3_12
J3_14
J3_16
J3_18
J3_20
J3_22
J3_24
J3_26
J3_28
J3_30
J3_32
J3_34
J3_36
J3_38
J3_40
J3_42
J3_44
GND
stm32h750_SPI_MOSI
stm32h750_SPI_SCK
AD9_3v
AD7_3v
LCD_DE-FPC_PIN9
AD5_3v
AD3_3v
AD1_3v
A19_latch_3v
A18_latch_3v
A16_latch_3v
A14_latch_3v
A13_latch_3v
A11_latch_3v
A9_latch_3v
A7_latch_3v
A5_latch_3v
A3_latch_3v
A1_latch_3v
GND
stm32h750_SPI_MISO
stm32h750_NOR_FLASH_CS
AD8_3v
LCD_PCLK-FPC_PIN37
AD6_3v
AD4_3v
AD2_3v
AD0_3v
stm32h750_buzzer
A17_latch_3v
A15_latch_3v
LCD_VSYNC-FPC_PIN10
A12_latch_3v
A10_latch_3v
A8_latch_3v
A6_latch_3v
A4_latch_3v
A2_latch_3v
A0_latch_3v
Module DevEBox STM32H7XX_M
V2.0
SN: 1907
Module DevEBox STM32H7XX_M
No es necesario conectar J4, se toman las señales del conector J3.
M4C
J4
3.3V
GND
PB15/MOSI
PB13/SCK
PB12/CS
PB14/MISO
PB1/RS
PB0/BLK
J4_1
J4_2
J4_3
J4_4
J4_5
J4_6
J4_7
J4_8
Module DevEBox STM32H7XX_M
V2.0
SN: 1907
Module DevEBox STM32H7XX_M
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Date: 10/17/2026
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Drawn By:

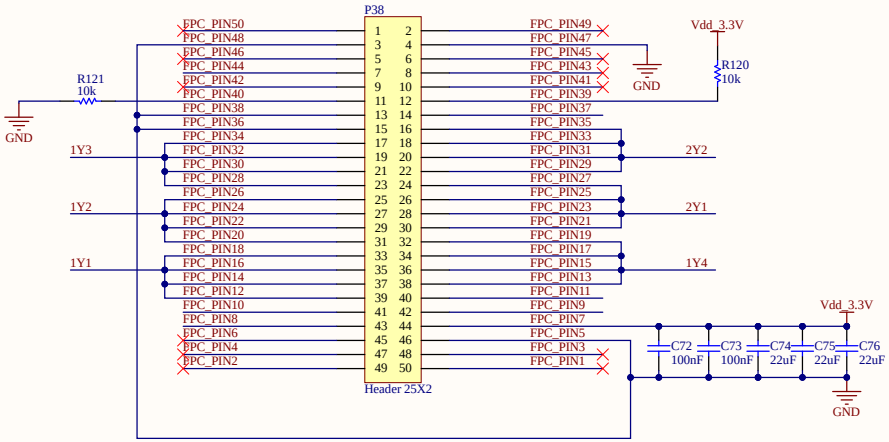
Jumpers para seleccionar si conectar el display LCD al MCU o a la FPGA.

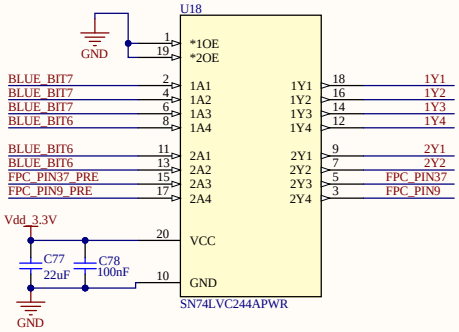


Definimos el pin 1 del zócalo FPC como el primer pin de la izquierda, mirando de frente la boca del zócalo.

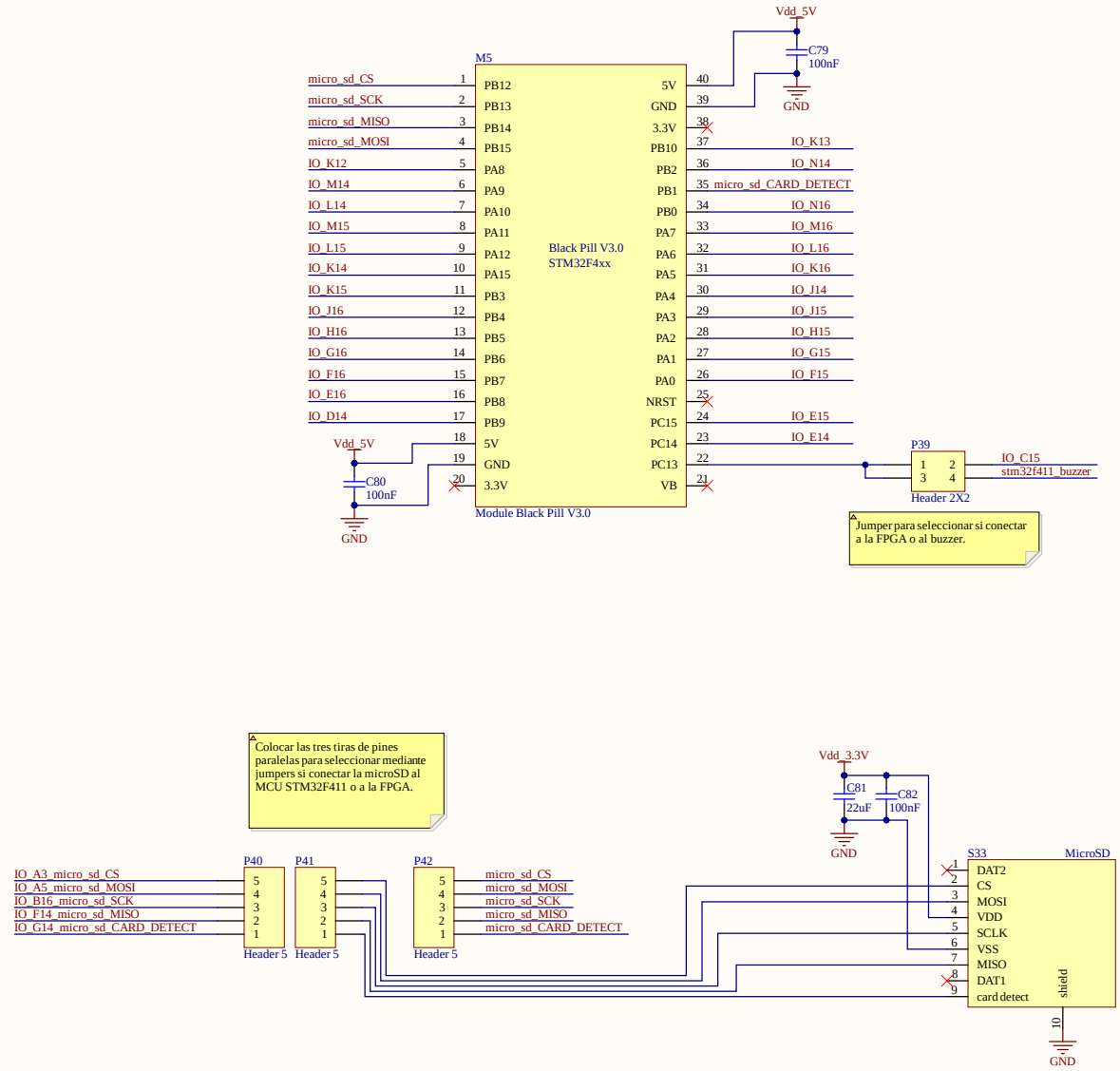
Esta es la convención utilizada en el datasheet de la pantalla, y las mediciones realizadas sobre el cable plano lo confirman.

Por lo tanto, el pin 1 del zócalo FPC corresponde al pin 50 del header 25x2, ya que el adaptador tiene invertida la numeración de los pines.





Title		
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Date: 10/17/2026	Sheet of	
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Title		
Size: A3	Number	Revision
Date: 10/17/2026	Sheet of	
File: C:\Users\..\stm32f411_dev_board.SchDoc	Drawn By:	